

Module Breakdown: Dedekind Domains and Ramification

Concept-level teaching package

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Assumptions and intake settings

- **Module scope:** Dedekind domains (integrality, unique factorization of ideals, norms and discriminants) and ramification of rational primes in number fields, including the Galois-theoretic side (decomposition groups, inertia groups, Frobenius elements).
- **Session count:** you answered “7–8”. The plan uses 7 core sessions plus an optional Session 8 (Frobenius in cyclotomic fields and quadratic reciprocity). If you only have 7 sessions, drop Session 8; nothing earlier depends on it.
- **Question bank type:** numerical (explicit computations in specific rings and fields).
- **Tool policy:** no calculators. All numbers are hand-sized: primes below 30 and residue computations modulo small primes.
- **Topic emphasis:** spread evenly across all topics.
- **Galois side:** included in full (decomposition and inertia groups, Frobenius).
- **Assumed audience (not specified; assumed):** first-year graduate students or strong advanced undergraduates who already know Galois theory (including finite fields), basic commutative algebra (Noetherian rings, localization, prime and maximal ideals), and elementary number theory (Legendre symbol, Euler’s criterion).

- **Conventions (assumed):**
 - A Dedekind domain is never a field.
 - Base field is always $\mathbb{Q}$; relative extensions appear only in towers.
 - The facts $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{2})} = \mathbb{Z}[\sqrt[3]{2}]$ and $\mathcal{O}_{\mathbb{Q}(\zeta_m)} = \mathbb{Z}[\zeta_m]$ are quoted without proof.
- **Question bank format:** no question in the bank was judged to benefit from a graph or diagram, so none is included. Visual aids appear only as instructor notes.

1 Topics and session plan

Session	Topic	Core content
1	T1. Integrality and Dedekind domains	Integral elements, integral closure, $\mathcal{O}_K$, rings of integers of quadratic fields, the definition of a Dedekind domain, examples and non-examples.
2	T2. Fractional ideals and unique factorization	Fractional ideals, invertibility, unique factorization into prime ideals, “to contain is to divide”, localization at primes (DVRs), the class group.
3	T3. Norms, discriminants, integral bases	Absolute ideal norm, discriminant d_K , the index formula, quadratic fields, the Minkowski bound as a computational tool.
4	T4. Primes in extensions: e , f , g	Lying over, ramification index, residue degree, the fundamental identity $\sum e_i f_i = n$, splitting types.
5	T5. The Dedekind–Kummer theorem	Factoring $p\mathcal{O}_K$ from $f \bmod p$, the quadratic splitting law, pure cubic examples, cyclotomic splitting.
6	T6. Ramification and the discriminant; orders	Dedekind’s discriminant theorem, index primes where Dedekind–Kummer fails, non-maximal orders and non-invertible ideals.
7	T7. Galois extensions: decomposition, inertia, Frobenius	Transitivity, $efg = n$, D and I , the exact sequence, Frobenius elements, dependence on the choice of $\mathfrak{P}$.
8 (optional)	T8. Frobenius in cyclotomic fields; quadratic reciprocity	$\text{Frob}_p = \sigma_p$ in $\mathbb{Q}(\zeta_m)$, splitting in subfields, $\mathbb{Q}(\sqrt{p^*}) \subset \mathbb{Q}(\zeta_p)$, a proof of quadratic reciprocity.

2 Topic-by-topic breakdown

2.1 T1. Integrality and Dedekind domains (Session 1)

Concepts

- Integral elements over a ring. Equivalently, b is integral over A if and only if $A[b]$ is a finitely generated A -module.
- Integral elements form a subring (the integral closure), and integrality is transitive.
- UFDs are integrally closed. In particular, the integral closure of $\mathbb{Z}$ in $\mathbb{Q}$ is $\mathbb{Z}$ (rational root theorem).

- An algebraic number is integral if and only if its minimal polynomial over $\mathbb{Q}$ has integer coefficients. For a quadratic irrationality this means: integral if and only if its trace and norm lie in $\mathbb{Z}$.
- The ring of integers $\mathcal{O}_K$ of a number field. It is a free $\mathbb{Z}$ -module of rank $n = [K : \mathbb{Q}]$ (statement now; proof in T3).
- $\mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{d})$ with d squarefree: it equals $\mathbb{Z}[\sqrt{d}]$ if $d \equiv 2, 3 \pmod{4}$, and $\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ if $d \equiv 1 \pmod{4}$.
- The three Dedekind axioms (Noetherian, integrally closed, every nonzero prime is maximal) and what each one rules out.
- Examples: $\mathbb{Z}$, $k[x]$, and every $\mathcal{O}_K$. For $\mathcal{O}_K$:
 - it is Noetherian because it is a finitely generated $\mathbb{Z}$ -module;
 - it is integrally closed by construction;
 - every nonzero prime is maximal because a finite integral domain is a field.
- Non-examples:
 - $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed;
 - $\mathbb{Z}[x]$ and $k[x, y]$ have Krull dimension 2;
 - the ring $\overline{\mathbb{Z}}$ of all algebraic integers is not Noetherian.

Key Definitions

- **Integral element.** Let $A \subseteq B$ be rings. An element $b \in B$ is *integral over* A if it is a root of a *monic* polynomial with coefficients in A . (Stress “monic”: every algebraic number is a root of some polynomial with integer coefficients.)
- **Integrally closed domain.** An integral domain A is *integrally closed* if every element of its fraction field $\text{Frac}(A)$ that is integral over A already lies in A . (The test elements come from $\text{Frac}(A)$, not from some larger field.)
- **Ring of integers.** For a number field K , $\mathcal{O}_K$ is the set of elements of K that are integral over $\mathbb{Z}$.
- **Dedekind domain.** An integral domain, not a field, that satisfies all three of the following:
 - it is Noetherian;
 - it is integrally closed;
 - every nonzero prime ideal is maximal.

Learning objectives

Students will be able to:

- **determine** whether a given algebraic number is integral by computing its minimal polynomial;
- **derive** $\mathcal{O}_K$ and an integral basis for every quadratic field;

- **explain** the role of each Dedekind axiom by giving, for each one, a ring that fails exactly that axiom;
- **classify** given rings as Dedekind or not, with justification.

Prerequisites

- Prime and maximal ideals; Noetherian rings (assumed background: commutative algebra).
- Number fields, minimal polynomials, degree of an extension (assumed background: field theory).
- Gauss's lemma and the rational root theorem (assumed background).
- Structure of finitely generated modules over a PID (assumed background).

Example questions with solutions

Q1.1 Let $\varphi = \frac{1+\sqrt{5}}{2}$.

- Show that φ is integral over $\mathbb{Z}$.
- Deduce that $\mathbb{Z}[\sqrt{5}]$ is not integrally closed.
- Find $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$.

Solution.

- We have $\varphi^2 = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} = \varphi + 1$. So φ is a root of the monic polynomial $x^2 - x - 1 \in \mathbb{Z}[x]$.
- φ lies in $\mathbb{Q}(\sqrt{5}) = \text{Frac}(\mathbb{Z}[\sqrt{5}])$ and is integral over $\mathbb{Z}$, hence integral over $\mathbb{Z}[\sqrt{5}]$. But $\varphi \notin \mathbb{Z}[\sqrt{5}]$, because its coefficients $\frac{1}{2}, \frac{1}{2}$ are not integers. So $\mathbb{Z}[\sqrt{5}]$ is not integrally closed.
- Since $5 \equiv 1 \pmod{4}$, the quadratic classification gives $\mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathbb{Z}[\varphi]$.

Q1.2 Decide which of the following rings are Dedekind domains, and for each non-Dedekind ring name the axiom that fails:

$$(a) \mathbb{Z}[\sqrt{-3}], \quad (b) \mathbb{Z}[x], \quad (c) \mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right], \quad (d) \overline{\mathbb{Z}}.$$

Solution.

- Not Dedekind: it is not integrally closed. The element $\omega = \frac{1+\sqrt{-3}}{2}$ satisfies $\omega^2 - \omega + 1 = 0$ and lies in $\mathbb{Q}(\sqrt{-3})$, but $\omega \notin \mathbb{Z}[\sqrt{-3}]$.
- Not Dedekind: $(0) \subsetneq (x) \subsetneq (2, x)$ is a chain of primes, so (x) is a nonzero prime that is not maximal. (The ring is Noetherian and integrally closed, since it is a UFD.)
- Dedekind: $-3 \equiv 1 \pmod{4}$, so this ring is the full ring of integers of $\mathbb{Q}(\sqrt{-3})$.
- Not Dedekind: it is not Noetherian. The chain $(2) \subsetneq (2^{1/2}) \subsetneq (2^{1/4}) \subsetneq \dots$ never stabilizes.
 - Each inclusion holds because $2^{1/2^k} = (2^{1/2^{k+1}})^2$.
 - Each inclusion is strict because $2^{1/2^{k+1}}$ is not a unit: its inverse has minimal polynomial $x^{2^{k+1}} - \frac{1}{2} \notin \mathbb{Z}[x]$, so the inverse is not integral.

Note that $\overline{\mathbb{Z}}$ is integrally closed and every nonzero prime is maximal. Only the Noetherian axiom fails.

Q1.3 Derive $\mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{d})$, where $d \neq 0, 1$ is squarefree.

Solution.

- Let $\alpha = a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$ and $b \neq 0$. Its minimal polynomial is $x^2 - 2ax + (a^2 - db^2)$, so α is integral if and only if $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$.
- Put $A = 2a$. Then $A^2 - d(2b)^2 = 4(a^2 - db^2) \in 4\mathbb{Z}$, so $d(2b)^2 \in \mathbb{Z}$.
- Since d is squarefree, $2b = B \in \mathbb{Z}$. (If $2b = r/s$ in lowest terms, then $s^2 \mid d$, which forces $s = 1$.)
- The condition becomes $A^2 \equiv dB^2 \pmod{4}$, where squares mod 4 are 0 or 1.
- If $d \equiv 2, 3 \pmod{4}$: B odd would force $A^2 \equiv 2$ or $3 \pmod{4}$, which is impossible. So B is even, then A is even, and $\alpha \in \mathbb{Z}[\sqrt{d}]$.
- If $d \equiv 1 \pmod{4}$: the condition is exactly $A \equiv B \pmod{2}$. So $\alpha = \frac{A+B\sqrt{d}}{2}$ with $A \equiv B \pmod{2}$, and these are precisely the elements of $\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$.

Instructor note — visual aid (sketch on the board)

(1) Where Dedekind domains sit among domains.

- Draw a large rectangle labeled “integral domains”.
- Inside it, draw two overlapping ovals: the left labeled “UFDs”, the right labeled “Dedekind domains”.
- Label the overlap “PIDs” and place $\mathbb{Z}$ and $\mathbb{Q}[x]$ there.
- In the UFD-only region, place $\mathbb{Z}[x]$ and $k[x, y]$.
- In the Dedekind-only region, place $\mathbb{Z}[\sqrt{-5}]$.
- Outside both ovals, place $\mathbb{Z}[\sqrt{-3}]$. Note beside it: “not integrally closed; $4 = 2 \cdot 2 = (1 + \sqrt{-3})(1 - \sqrt{-3})$ ”.
- The point to make: for Dedekind domains, “UFD” and “PID” coincide.

(2) $\mathbb{Z}[\sqrt{5}]$ inside $\mathbb{Z}[\varphi]$ as lattices.

- Use axes labeled “rational part” (horizontal) and “coefficient of $\sqrt{5}$ ” (vertical), both from -2 to 2 .
- Mark the points of $\mathbb{Z}[\sqrt{5}]$ (integer coordinates) as filled dots.
- Mark the additional points of $\mathbb{Z}[\varphi]$ (both coordinates half-odd-integers, such as $(\frac{1}{2}, \frac{1}{2})$) as hollow dots.
- Label φ at $(\frac{1}{2}, \frac{1}{2})$.
- The filled dots form an index-2 sublattice of all the dots: $\mathbb{Z}[\sqrt{5}]$ misses exactly the points at the “centers” of its unit squares.

2.2 T2. Fractional ideals and unique factorization (Session 2)

Concepts

- Fractional ideals, their products, and principal fractional ideals.
- Invertibility. The candidate inverse is always $(A : I) = \{x \in K : xI \subseteq A\}$.
- The main theorem: in a Dedekind domain A ,
 - every nonzero fractional ideal is invertible;
 - every nonzero ideal factors uniquely into prime ideals;
 - the fractional ideals form the free abelian group on the nonzero primes.
- Proof ingredients:
 - every nonzero ideal contains a product of nonzero primes;
 - for a nonzero prime $\mathfrak{p}$, $\mathfrak{p}^{-1} \not\supseteq A$;
 - hence $\mathfrak{p}\mathfrak{p}^{-1} = A$ by maximality.
- “To contain is to divide.” Valuations $v_{\mathfrak{p}}$. The gcd of two ideals is their sum; the lcm is their intersection.
- Local characterization: a Noetherian domain (not a field) is Dedekind if and only if every localization at a nonzero prime is a DVR.
- The class group $\text{Cl}(A)$. For Dedekind A : A is a PID $\iff A$ is a UFD $\iff \text{Cl}(A) = 1$.
- Elements versus ideals: irreducible elements need not generate prime ideals. Non-unique element factorizations are reconciled at the level of ideals.
- Every ideal of a Dedekind domain is generated by two elements (statement).

Key Definitions

- **Fractional ideal.** Let A be a domain with fraction field K . A fractional ideal is a *nonzero* A -submodule $I \subseteq K$ such that $dI \subseteq A$ for some nonzero $d \in A$. (So it has a common denominator.)
- **Invertible fractional ideal.** I is invertible if $IJ = A$ for some fractional ideal J . When this happens, necessarily $J = I^{-1} = \{x \in K : xI \subseteq A\}$.
- **Divisibility of ideals.** $\mathfrak{a} \mid \mathfrak{b}$ means $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$ for some ideal $\mathfrak{c}$. In a Dedekind domain this holds *if and only if* $\mathfrak{a} \supseteq \mathfrak{b}$: the bigger ideal is the divisor.
- **$\mathfrak{p}$ -adic valuation of an ideal.** $v_{\mathfrak{p}}(\mathfrak{a})$ is the exponent of $\mathfrak{p}$ in the prime factorization of $\mathfrak{a}$.
- **Ideal class group.** $\text{Cl}(A)$ is the group of fractional ideals modulo the subgroup of nonzero principal fractional ideals. Its order h is the class number.

Learning objectives

Students will be able to:

- **compute** products and inverses of explicit ideals in quadratic rings;

- **factor** principal ideals into prime ideals and **explain** how non-unique element factorizations become a single ideal factorization;
- **apply** “to contain is to divide” to compute gcds and lcms of ideals;
- **explain** why invertibility fails in a ring that is not integrally closed, using an explicit ideal.

Prerequisites

- Definition of a Dedekind domain and quadratic rings of integers (earlier in this module: T1).
- Localization and local rings (assumed background).
- Noetherian induction (assumed background).

Example questions with solutions

Throughout, let $\mathcal{O} = \mathbb{Z}[\sqrt{-5}]$, and write

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \mathfrak{p}'_3 = (3, 1 - \sqrt{-5}).$$

Q2.1 Show that $(2) = \mathfrak{p}_2^2$, $(3) = \mathfrak{p}_3\mathfrak{p}'_3$, $(1 + \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}_3$ and $(1 - \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}'_3$. Deduce that the two factorizations $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ give the same ideal factorization.

Solution.

- $\mathfrak{p}_2^2 = (4, 2 + 2\sqrt{-5}, (1 + \sqrt{-5})^2)$, and $(1 + \sqrt{-5})^2 = -4 + 2\sqrt{-5}$. Every generator is divisible by 2. The difference $(2 + 2\sqrt{-5}) - (-4 + 2\sqrt{-5}) = 6$ lies in $\mathfrak{p}_2^2$, and so does 4. Hence $2 = 6 - 4 \in \mathfrak{p}_2^2$, and $\mathfrak{p}_2^2 = (2)$.
- $\mathfrak{p}_3\mathfrak{p}'_3 = (9, 3(1 - \sqrt{-5}), 3(1 + \sqrt{-5}), 6)$. Every generator is divisible by 3, and $3 = 9 - 6$ lies in the ideal. Hence $\mathfrak{p}_3\mathfrak{p}'_3 = (3)$.
- $\mathfrak{p}_2\mathfrak{p}_3 = (6, 2 + 2\sqrt{-5}, 3 + 3\sqrt{-5}, -4 + 2\sqrt{-5})$. It contains $(3 + 3\sqrt{-5}) - (2 + 2\sqrt{-5}) = 1 + \sqrt{-5}$. Conversely, every generator is a multiple of $1 + \sqrt{-5}$, because $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ and $-4 + 2\sqrt{-5} = (1 + \sqrt{-5})^2$. Hence $\mathfrak{p}_2\mathfrak{p}_3 = (1 + \sqrt{-5})$. The case $\mathfrak{p}_2\mathfrak{p}'_3 = (1 - \sqrt{-5})$ is the same computation with $\sqrt{-5}$ replaced by $-\sqrt{-5}$.
- Therefore

$$(6) = (2)(3) = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{p}'_3 \quad \text{and} \quad (6) = (1 + \sqrt{-5})(1 - \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}_3 \cdot \mathfrak{p}_2\mathfrak{p}'_3 = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{p}'_3.$$

The two element factorizations just group the same four prime ideals differently.

Q2.2 Compute $\mathfrak{p}_2^{-1}$ explicitly and check that $\mathfrak{p}_2\mathfrak{p}_2^{-1} = \mathcal{O}$. Show that $\mathfrak{p}_2^{-1} \not\subseteq \mathcal{O}$.

Solution.

- From $\mathfrak{p}_2^2 = (2)$ we get $\mathfrak{p}_2 \cdot \frac{1}{2}\mathfrak{p}_2 = \frac{1}{2}(2) = \mathcal{O}$. So $\mathfrak{p}_2^{-1} = \frac{1}{2}\mathfrak{p}_2 = \left\{ \frac{a+b\sqrt{-5}}{2} : a \equiv b \pmod{2} \right\}$.
- Check against the formula $\{x : x\mathfrak{p}_2 \subseteq \mathcal{O}\}$ with $x = \frac{1+\sqrt{-5}}{2}$:

$$2x = 1 + \sqrt{-5} \in \mathcal{O}, \quad x(1 + \sqrt{-5}) = \frac{-4+2\sqrt{-5}}{2} = -2 + \sqrt{-5} \in \mathcal{O}.$$

- This same x is not in $\mathcal{O}$, so $\mathfrak{p}_2^{-1} \not\subseteq \mathcal{O}$.

Q2.3 In $R = \mathbb{Z}[\sqrt{-3}]$, let $P = (2, 1 + \sqrt{-3})$. Show that $P^2 = 2P$ but $P \neq (2)$. Conclude that P is not invertible, so R is not Dedekind.

Solution.

- $(1 + \sqrt{-3})^2 = -2 + 2\sqrt{-3}$. So

$$P^2 = (4, 2 + 2\sqrt{-3}, -2 + 2\sqrt{-3}) = (4, 2 + 2\sqrt{-3}),$$

since the third generator equals the second minus 4.

- Also $2P = (4, 2 + 2\sqrt{-3})$. Hence $P^2 = 2P$.
- If P were invertible, multiplying $P^2 = 2P$ by P^{-1} would give $P = (2)$. But $1 + \sqrt{-3} \in P$ is not divisible by 2 in R . So P is not invertible.

Instructor note — visual aid (sketch on the board)

(1) The ideal $\mathfrak{p}_2$ as a checkerboard sublattice.

- Draw axes: horizontal “ a ” and vertical “ b ” for $a + b\sqrt{-5}$, each from -3 to 3 .
- Mark all integer points (a, b) as small dots: this is $\mathcal{O}$.
- Circle those with $a \equiv b \pmod{2}$: this is $\mathfrak{p}_2$. It forms a checkerboard pattern.
- Label 2 at $(2, 0)$ and $1 + \sqrt{-5}$ at $(1, 1)$ as the two lattice generators, and shade the parallelogram they span.
- Its area is twice that of the unit square, which makes $[\mathcal{O} : \mathfrak{p}_2] = 2$ visible.
- Repeat beside it for $\mathfrak{p}_3 = \{a \equiv b \pmod{3}\}$: circle every third point along each diagonal band.

(2) Divisors of (6) as a Hasse diagram.

- Put $\mathcal{O}$ at the top.
- Next row: the primes $\mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}'_3$.
- Next row: $(2) = \mathfrak{p}_2^2$, $(1 + \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}_3$, $(1 - \sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}'_3$, $(3) = \mathfrak{p}_3\mathfrak{p}'_3$, and the other products as needed.
- Put (6) at the bottom.
- Draw an edge whenever one ideal contains the next. Write in the margin: “higher = bigger ideal = divisor”.

2.3 T3. Norms, discriminants, integral bases (Session 3)

Concepts

- The absolute norm $N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|$, which is finite.
- Multiplicativity $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$, via the Chinese remainder theorem and $|\mathfrak{p}^k/\mathfrak{p}^{k+1}| = |\mathcal{O}_K/\mathfrak{p}|$.
- $N(\alpha\mathcal{O}_K) = |N_{K/\mathbb{Q}}(\alpha)|$.
- $N(\mathfrak{p}) = p^f$, where f is the residue degree (named properly in T4).

- Discriminant of an n -tuple: $\det(\sigma_i(\alpha_j))^2 = \det(\text{Tr}(\alpha_i \alpha_j))$. Under a change of basis by M it changes by $\det(M)^2$.
- $\mathcal{O}_K$ is free of rank n ; integral bases; d_K is independent of the basis chosen.
- The index formula $\text{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 d_K$. Consequence: if $\text{disc}(f)$ is squarefree, then $\mathcal{O}_K = \mathbb{Z}[\alpha]$.
- Polynomial discriminants:
 - $\text{disc}(f) = (-1)^{n(n-1)/2} N_{K/\mathbb{Q}}(f'(\alpha))$;
 - $x^2 + bx + c$ has discriminant $b^2 - 4c$;
 - $x^3 + ax + b$ has discriminant $-4a^3 - 27b^2$.
- Quadratic fields: $d_K = d$ if $d \equiv 1 \pmod{4}$, and $d_K = 4d$ otherwise.
- Minkowski bound (stated, not proved): every ideal class contains an integral ideal of norm at most

$$M_K = \frac{n!}{n^n} \left(\frac{4}{\pi} \right)^{r_2} \sqrt{|d_K|}.$$

Key Definitions

- **Absolute norm of an ideal.** For a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$, $N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|$, the number of elements of the quotient ring. (It is a count of residues, not a norm of a generator. The two agree, up to sign, only for principal ideals.)
- **Discriminant of a basis.** For $\alpha_1, \dots, \alpha_n \in K$, $\text{disc}(\alpha_1, \dots, \alpha_n) = \det(\sigma_i(\alpha_j))^2 = \det(\text{Tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j))$, where the σ_i are the n embeddings $K \hookrightarrow \mathbb{C}$.
- **Integral basis; discriminant of K .** An integral basis is a $\mathbb{Z}$ -basis of $\mathcal{O}_K$. The discriminant d_K is the discriminant of any integral basis.
- **Index.** For $\alpha \in \mathcal{O}_K$ generating K , the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is the (finite) group index of $\mathbb{Z}[\alpha]$ in $\mathcal{O}_K$.

Learning objectives

Students will be able to:

- **compute** norms of explicit ideals by identifying the quotient ring;
- **compute** discriminants of quadratic orders and trinomial cubic orders by hand;
- **apply** the index formula to decide whether $\mathcal{O}_K = \mathbb{Z}[\alpha]$;
- **apply** the Minkowski bound to determine the class number of $\mathbb{Q}(\sqrt{-5})$.

Prerequisites

- Trace and norm for finite field extensions; Vandermonde determinants (assumed background).
- The Chinese remainder theorem for rings; finitely generated abelian groups and lattices (assumed background).
- Unique factorization of ideals and the class group (earlier in this module: T2).

- Rings of integers of quadratic fields (earlier in this module: T1).

Example questions with solutions

Q3.1 Compute $\text{disc}(\mathbb{Z}[\sqrt{-5}])$ and $\text{disc}\left(\mathbb{Z}\left[\frac{1+\sqrt{13}}{2}\right]\right)$.

Solution.

- For the basis $(1, \sqrt{d})$:

$$\det \begin{pmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{d} \end{pmatrix}^2 = (-2\sqrt{d})^2 = 4d.$$

With $d = -5$ this gives -20 .

- For the basis $(1, \frac{1+\sqrt{d}}{2})$, the determinant is $\frac{1-\sqrt{d}}{2} - \frac{1+\sqrt{d}}{2} = -\sqrt{d}$, whose square is d . With $d = 13$ this gives 13 .

Q3.2 (a) Use the index formula to find $[\mathcal{O}_{\mathbb{Q}(\sqrt{13})} : \mathbb{Z}[\sqrt{13}]]$.

(b) Let α be a root of $x^3 - x - 1$. Show that $\mathcal{O}_{\mathbb{Q}(\alpha)} = \mathbb{Z}[\alpha]$.

Solution.

- (a) $\text{disc}(x^3 - x - 1) = 0 + 52 = 52$ and $d_K = 13$. So the index squared is $52/13 = 4$, and the index is 2 .
- (b) The polynomial has no rational roots (± 1 are not roots), so it is irreducible. Its discriminant is $-4(-1)^3 - 27(1)^2 = 4 - 27 = -23$. From $-23 = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 d_K$ and the fact that 23 is squarefree, the index is 1 .

Q3.3 In $K = \mathbb{Q}(\sqrt{-5})$:

- (a) Compute $N(\mathfrak{p}_2)$, $N(\mathfrak{p}_3)$ and $N((1 + \sqrt{-5}))$, and check multiplicativity.
- (b) Show that $\mathfrak{p}_2$ is not principal.
- (c) Use the Minkowski bound to prove $h_K = 2$.

Solution.

- (a) $\mathcal{O}/\mathfrak{p}_2 \cong \mathbb{Z}[x]/(2, 1+x, x^2+5) \cong \mathbb{F}_2[x]/(x+1) \cong \mathbb{F}_2$, since $x = -1$ gives $1+5 \equiv 0 \pmod{2}$. So $N(\mathfrak{p}_2) = 2$.

Similarly $\mathcal{O}/\mathfrak{p}_3 \cong \mathbb{F}_3$ (with $x = -1$, $1+5 = 6 \equiv 0 \pmod{3}$), so $N(\mathfrak{p}_3) = 3$.

Finally $N((1 + \sqrt{-5})) = |1+5| = 6 = 2 \cdot 3$, consistent with $(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3$.

- (b) If $\mathfrak{p}_2 = (a + b\sqrt{-5})$, then $a^2 + 5b^2 = 2$, which has no integer solutions.
- (c) Here $n = 2$, $r_2 = 1$ and $|d_K| = 20$, so $M_K = \frac{2}{4} \cdot \frac{4}{\pi} \sqrt{20} = \frac{2}{\pi} \sqrt{20}$. Since $4.5^2 = 20.25 > 20$ and $\pi > 3$,

$$M_K < \frac{2 \cdot 4.5}{\pi} = \frac{9}{\pi} < 3.$$

So every class contains an ideal of norm 1 or 2 . The only such ideals are $\mathcal{O}$ and $\mathfrak{p}_2$ (the unique prime above 2). Since $\mathfrak{p}_2$ is not principal, $h_K = 2$.

Instructor note — visual aid (sketch on the board)

Covolume picture: discriminant as squared area.

- Draw the complex plane with $\mathbb{Z}[i]$ marked as the integer grid and shade the unit square spanned by 1 and i . Label it “area = $1 = \frac{1}{2}\sqrt{|d_K|}$, $d_K = -4$ ”.
- Beside it, draw $\mathbb{Z}[\sqrt{-5}]$: points $a + b\sqrt{-5}$ plotted at $(a, b\sqrt{5})$, a rectangular grid with vertical spacing about 2.24. Shade the $1 \times \sqrt{5}$ rectangle and label it “area = $\sqrt{5} = \frac{1}{2}\sqrt{20}$ ”.
- State the general rule under the pictures: the covolume of $\mathcal{O}_K$ in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ is $2^{-r_2}\sqrt{|d_K|}$.
- Point out that a sublattice of index k has covolume k times bigger. This is the geometric reason behind $\text{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 d_K$.

2.4 T4. Primes in extensions: e, f, g (Session 4)

Concepts

- Lying over: each nonzero prime $\mathfrak{P}$ of $\mathcal{O}_K$ contains exactly one rational prime p , and $\mathfrak{P} \supseteq p\mathcal{O}_K$. Every p has at least one prime above it.
- The factorization $p\mathcal{O}_K = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_g^{e_g}$. The residue fields $\mathcal{O}_K/\mathfrak{P}_i$ are $\mathbb{F}_{p^{f_i}}$.
- The fundamental identity $\sum_{i=1}^g e_i f_i = n$. Proof: by CRT, $\mathcal{O}_K/p\mathcal{O}_K \cong \prod_i \mathcal{O}_K/\mathfrak{P}_i^{e_i}$. The left side has p^n elements and the i -th factor has $p^{e_i f_i}$.
- Vocabulary: split completely ($g = n$), inert ($g = 1, f = n$), ramified (some $e_i > 1$), totally ramified ($e = n$), splitting type.
- Reading e and f off the ring: $\mathcal{O}_K/p\mathcal{O}_K$ has nonzero nilpotents if and only if p ramifies.
- Multiplicativity in towers: $e(\mathfrak{q} | p) = e(\mathfrak{q} | \mathfrak{P}) e(\mathfrak{P} | p)$, and likewise for f .
- Possible splitting types:
 - quadratic fields: 3 types;
 - cubic fields: 5 types.

Key Definitions

- **Lying over.** A prime $\mathfrak{P}$ of $\mathcal{O}_K$ *lies over* p (written $\mathfrak{P} | p$) if $\mathfrak{P} \cap \mathbb{Z} = p\mathbb{Z}$. Equivalently, $\mathfrak{P} \supseteq p\mathcal{O}_K$.
- **Ramification index.** $e(\mathfrak{P} | p) = v_{\mathfrak{P}}(p\mathcal{O}_K)$, the exponent of $\mathfrak{P}$ in the factorization of $p\mathcal{O}_K$.
- **Residue (inertia) degree.** $f(\mathfrak{P} | p) = [\mathcal{O}_K/\mathfrak{P} : \mathbb{F}_p]$, so that $N(\mathfrak{P}) = p^f$. It is a degree of *fields*; it has nothing to do with multiplicity.
- **Ramified.** p ramifies in K if $e(\mathfrak{P} | p) > 1$ for *some* $\mathfrak{P} | p$. It need not hold for all of them.
- **Split completely / inert / totally ramified.** Split completely: $p\mathcal{O}_K$ is a product of n distinct primes (all $e = f = 1$). Inert: $p\mathcal{O}_K$ is itself prime ($f = n$). Totally ramified:

$$p\mathcal{O}_K = \mathfrak{P}^n.$$

Learning objectives

Students will be able to:

- **identify** e, f, g from a factorization or from the structure of $\mathcal{O}_K/p\mathcal{O}_K$;
- **derive** the fundamental identity from CRT and ideal norms;
- **enumerate** all possible splitting types in degree 2 and 3;
- **explain** why ramification corresponds to nilpotents in $\mathcal{O}_K/p\mathcal{O}_K$.

Prerequisites

- Unique factorization of ideals (earlier in this module: T2).
- Ideal norms and $N(\mathfrak{P}) = p^f$ (earlier in this module: T3).
- Finite fields and their extensions (assumed background).
- The Chinese remainder theorem (assumed background).

Example questions with solutions

Q4.1 In $\mathbb{Z}[i]$, factor 2, 3 and 5. Give e, f, g in each case and verify $\sum e_i f_i = 2$.

Solution. Since $\mathbb{Z}[i] \cong \mathbb{Z}[x]/(x^2 + 1)$, we have $\mathbb{Z}[i]/(p) \cong \mathbb{F}_p[x]/(x^2 + 1)$.

- $p = 2$: $x^2 + 1 = (x + 1)^2$ in $\mathbb{F}_2[x]$, so $(2) = (1 + i)^2$. Here $e = 2, f = 1, g = 1$, and $2 \cdot 1 = 2$.
- $p = 3$: $x^2 + 1$ has no roots mod 3, so $\mathbb{F}_3[x]/(x^2 + 1) = \mathbb{F}_9$ and (3) is prime. Here $e = 1, f = 2, g = 1$.
- $p = 5$: $x^2 + 1 = (x - 2)(x + 2)$ in $\mathbb{F}_5[x]$, so $(5) = (5, i - 2)(5, i + 2)$. Since $N(i - 2) = 5$, the ideal $(i - 2)$ is contained in $(5, i - 2)$ and has the same norm, so $(5, i - 2) = (i - 2)$. Thus $(5) = (2 + i)(2 - i)$ with $e = f = 1, g = 2$, and $1 + 1 = 2$.

Q4.2 Let K be a cubic field. What does each of the following tell you about the factorization of p ?

- $\mathcal{O}_K/3\mathcal{O}_K \cong \mathbb{F}_3 \times \mathbb{F}_9$
- $\mathcal{O}_K/7\mathcal{O}_K \cong \mathbb{F}_7[t]/(t^2) \times \mathbb{F}_7$

Solution. Match each factor with $\mathcal{O}_K/\mathfrak{P}_i^{e_i}$. The residue field of a factor gives f_i , and the nilpotency length gives e_i .

- Both factors are fields, so $(3) = \mathfrak{P}_1 \mathfrak{P}_2$ with $f_1 = 1, f_2 = 2$ and all $e_i = 1$. Check: $1 + 2 = 3$.
- The factor $\mathbb{F}_7[t]/(t^2)$ has residue field $\mathbb{F}_7$ and t nilpotent of order 2, so $e_1 = 2, f_1 = 1$. The second factor gives $e_2 = f_2 = 1$. So $(7) = \mathfrak{P}_1^2 \mathfrak{P}_2$, and $2 + 1 = 3$.

Q4.3 List all possible splitting types of a prime p in a cubic field.

Solution. We need the multisets of pairs (e_i, f_i) with $\sum e_i f_i = 3$:

- $\{(1, 1), (1, 1), (1, 1)\}$: split completely;
- $\{(1, 1), (1, 2)\}$;
- $\{(1, 3)\}$: inert;
- $\{(2, 1), (1, 1)\}$: partially ramified;
- $\{(3, 1)\}$: totally ramified.

These are five types. (The Galois case in T7 will rule some of them out.)

Instructor note — visual aid (sketch on the board)

“Primes of $\mathbb{Z}[i]$ over primes of $\mathbb{Z}$ ” picture.

- Draw a horizontal line near the bottom of the board labeled “primes of $\mathbb{Z}$ ”. Mark dots at 2, 3, 5, 7, 11, 13, evenly spaced.
- Above it, draw a wavy horizontal curve labeled “primes of $\mathbb{Z}[i]$ ”. Above each p , draw its primes as follows:
 - over 2: one dot, with the curve drawn touching the line there like a tangency; label “ $e = 2$ ”;
 - over 3, 7 and 11: one large hollow dot each; label “ $f = 2$, inert”;
 - over 5 and 13: two separate small dots each (the curve crosses the vertical above p twice); label “split”.
- Draw thin vertical dashed lines connecting each prime of $\mathbb{Z}$ to the primes above it.
- Write the rule in the corner: “ $p \equiv 1 \pmod{4}$: split; $p \equiv 3 \pmod{4}$: inert; $p = 2$: ramified”.
- The point to emphasize: e is a multiplicity (tangency), f is the size of the residue field (a big dot), and g is a count (how many dots).

2.5 T5. The Dedekind–Kummer theorem (Session 5)

Concepts

- Statement of Dedekind–Kummer, and the proof idea: if $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$, then $\mathcal{O}_K/p\mathcal{O}_K \cong \mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \cong \mathbb{F}_p[x]/(\bar{f})$.
- Why the index hypothesis matters (developed in T6).
- Explicit two-element generators $\mathfrak{P}_i = (p, g_i(\alpha))$.
- The quadratic splitting law via Legendre symbols, including the special rule at $p = 2$.
- The pure cubic $\mathbb{Q}(\sqrt[3]{2})$: splitting via cubic residues.
- Cyclotomic fields (with $\mathcal{O}_K = \mathbb{Z}[\zeta_m]$ quoted):
 - for $p \nmid m$: Φ_m factors mod p into $\varphi(m)/f$ irreducibles of degree $f = \text{ord}_m(p)$;
 - p is totally ramified in $\mathbb{Q}(\zeta_{p^k})$.

Key Definitions

- **Dedekind–Kummer theorem (state it verbatim, hypothesis first).** Let $K = \mathbb{Q}(\alpha)$ with $\alpha \in \mathcal{O}_K$ and minimal polynomial $f \in \mathbb{Z}[x]$. Let p be a prime with $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Suppose

$$\bar{f} = \prod_{i=1}^g \bar{g}_i^{e_i} \quad \text{in } \mathbb{F}_p[x],$$

with the $\bar{g}_i$ distinct monic irreducibles, and let $g_i \in \mathbb{Z}[x]$ be any lifts. Then

$$p\mathcal{O}_K = \prod_{i=1}^g \mathfrak{P}_i^{e_i}, \quad \mathfrak{P}_i = (p, g_i(\alpha)), \quad f(\mathfrak{P}_i | p) = \deg \bar{g}_i.$$

- **Quadratic splitting law.** Let $K = \mathbb{Q}(\sqrt{d})$ with d squarefree.
 - Odd p : ramified if $p \mid d$; split if $\left(\frac{d}{p}\right) = 1$; inert if $\left(\frac{d}{p}\right) = -1$.
 - $p = 2$: ramified if $d \equiv 2, 3 \pmod{4}$; split if $d \equiv 1 \pmod{8}$; inert if $d \equiv 5 \pmod{8}$.
- **Multiplicative order.** For $p \nmid m$, $\text{ord}_m(p)$ is the least $k \geq 1$ with $p^k \equiv 1 \pmod{m}$.

Learning objectives

Students will be able to:

- **apply** Dedekind–Kummer to factor $p\mathcal{O}_K$ and write explicit generators;
- **derive** the quadratic splitting law from Dedekind–Kummer;
- **predict** splitting in $\mathbb{Q}(\zeta_m)$ from $\text{ord}_m(p)$;
- **determine** splitting in $\mathbb{Q}(\sqrt[3]{2})$ using cubes mod p .

Prerequisites

- The index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ (earlier in this module: T3).
- e, f, g and the fundamental identity (earlier in this module: T4).
- Factoring polynomials over $\mathbb{F}_p$; Legendre symbol; Euler’s criterion (assumed background).
- $\mathbb{F}_{p^f}^\times$ is cyclic of order $p^f - 1$ (assumed background).

Example questions with solutions

Q5.1 Factor 2, 3, 5, 7 and 11 in $\mathbb{Z}[\sqrt{-5}]$.

Solution. Here $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ (since $-5 \equiv 3 \pmod{4}$), so the index is 1. Take $f = x^2 + 5$.

- Mod 2: $x^2 + 1 = (x + 1)^2$, so $(2) = (2, 1 + \sqrt{-5})^2$. Ramified.
- Mod 3: $x^2 + 2 = x^2 - 1 = (x - 1)(x + 1)$, so $(3) = (3, \sqrt{-5} - 1)(3, \sqrt{-5} + 1)$. Split.
- Mod 5: $\bar{f} = x^2$, so $(5) = (5, \sqrt{-5})^2 = (\sqrt{-5})^2$. Ramified.

- Mod 7: $x^2 + 5 \equiv x^2 - 2$, and $3^2 = 9 \equiv 2 \pmod{7}$. So $\bar{f} = (x - 3)(x + 3)$ and $(7) = (7, \sqrt{-5} - 3)(7, \sqrt{-5} + 3)$. Split.
- Mod 11: $x^2 + 5 \equiv x^2 - 6$. The squares mod 11 are 1, 4, 9, 5, 3, so 6 is not a square. Hence $\bar{f}$ is irreducible and 11 is inert.

Q5.2 Let $\alpha = \sqrt[3]{2}$, so $\mathcal{O}_K = \mathbb{Z}[\alpha]$. Factor 2, 3, 5 and 7.

Solution. Take $f = x^3 - 2$.

- Mod 2: $\bar{f} = x^3$, so $(2) = (\alpha)^3$. Totally ramified.
- Mod 3: $x^3 - 2 \equiv x^3 + 1 = (x + 1)^3$ (Frobenius), so $(3) = (3, \alpha + 1)^3$. Totally ramified.
As a check: the minimal polynomial of $\alpha + 1$ is $(x - 1)^3 - 2$, whose constant term is -3 , so $N(\alpha + 1) = 3$. In fact $(3) = (\alpha + 1)^3$ as ideals, because $(\alpha + 1)^3 = 3(1 + \alpha + \alpha^2)$ and $1 + \alpha + \alpha^2 = 1/(\alpha - 1)$ is a unit.
- Mod 5: $3^3 = 27 \equiv 2$, so $x = 3$ is a root. Dividing, $\bar{f} = (x - 3)(x^2 + 3x + 4)$. Check:

$$(x - 3)(x^2 + 3x + 4) = x^3 - 5x - 12 \equiv x^3 - 2 \pmod{5}.$$

The quadratic has discriminant $9 - 16 = -7 \equiv 3 \pmod{5}$, which is not a square mod 5. So

$$(5) = \mathfrak{P}_1 \mathfrak{P}_2, \quad \mathfrak{P}_1 = (5, \alpha - 3), \quad f(\mathfrak{P}_1 | 5) = 1, \quad \mathfrak{P}_2 = (5, \alpha^2 + 3\alpha + 4), \quad f(\mathfrak{P}_2 | 5) = 2.$$

- Mod 7: the cubes mod 7 are 0, 1, 6, so 2 is not a cube. A cubic with no roots is irreducible, so 7 is inert.

Q5.3 In $\mathbb{Q}(\zeta_5)$ (with $\mathcal{O}_K = \mathbb{Z}[\zeta_5]$), determine the splitting of 2, 11, 19 and 5. For 11, find the four primes explicitly.

Solution. For $p \neq 5$: $f = \text{ord}_5(p)$ and $g = 4/f$.

- $p = 2$: 2, 4, 3, 1 are the successive powers, so $f = 4$ and 2 is inert.
- $p = 11$: $11 \equiv 1 \pmod{5}$, so $f = 1$ and 11 splits completely. The roots of Φ_5 mod 11 are the elements of order 5 in $\mathbb{F}_{11}^\times$. Since $3^5 = 243 = 242 + 1 \equiv 1$, the element 3 has order 5, with powers 3, 9, 5, 4. Hence

$$(11) = \prod_{a \in \{3, 4, 5, 9\}} (11, \zeta_5 - a).$$

- $p = 19$: $19 \equiv 4 \pmod{5}$ and $4^2 \equiv 1$, so $f = 2$ and $g = 2$.
- $p = 5$: $\Phi_5(x) = \frac{x^5 - 1}{x - 1} \equiv \frac{(x - 1)^5}{x - 1} = (x - 1)^4 \pmod{5}$. So $(5) = (5, \zeta_5 - 1)^4 = (1 - \zeta_5)^4$, totally ramified.

Instructor note — visual aid (sketch on the board)

Dictionary diagram: polynomial factors $\leftrightarrow$ prime ideals.

- Draw two columns. Head the left one “factors of $\bar{f}$ in $\mathbb{F}_p[x]$ ” and the right one “primes of $\mathcal{O}_K$ above p ”.
- First example: $\mathbb{Q}(\sqrt[3]{2})$ at $p = 5$.

- Left: $(x - 3)$ and $(x^2 + 3x + 4)$, each with its exponent 1 written as a superscript.
- Right: $(5, \alpha - 3)$ and $(5, \alpha^2 + 3\alpha + 4)$.
- Connect matching entries with arrows.
- Annotate the arrows with three rules: “exponent $\rightarrow e$ ”, “degree $\rightarrow f$ ”, “number of distinct factors $\rightarrow g$ ”.
- Second example underneath, same format: $\mathbb{Q}(\sqrt[3]{2})$ at $p = 3$, where the left column is $(x + 1)^3$ and the right column is $(3, \alpha + 1)^3$.
- Above both columns, draw a box around the hypothesis “ $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ ” with an arrow pointing to the whole diagram. This foreshadows T6.

2.6 T6. Ramification and the discriminant; orders (Session 6)

Concepts

- Dedekind’s discriminant theorem: p ramifies in K if and only if $p \mid d_K$. Consequences:
 - only finitely many primes ramify;
 - every $K \neq \mathbb{Q}$ has a ramified prime, because Minkowski’s bound forces $|d_K| > 1$ (statement).
- Proof idea: p ramifies $\iff \mathcal{O}_K/p\mathcal{O}_K$ has nonzero nilpotents $\iff$ the trace form on $\mathcal{O}_K/p\mathcal{O}_K$ over $\mathbb{F}_p$ is degenerate $\iff p \mid d_K$.
- $\text{disc}(f)$ versus d_K . They differ by the square of the index, so primes dividing the index can divide $\text{disc}(f)$ without ramifying.
- What to do at an index prime: use another generator, such as $\frac{1+\sqrt{d}}{2}$ instead of $\sqrt{d}$.
- Orders: subrings of finite index in $\mathcal{O}_K$.
 - A non-maximal order is not integrally closed, hence not Dedekind.
 - Ideals at primes dividing the conductor can fail to be invertible.
- (Optional enrichment) The different $\mathfrak{D}_K$: $|\mathcal{N}(\mathfrak{D}_K)| = |d_K|$, and $\mathfrak{P} \mid \mathfrak{D}_K$ if and only if $\mathfrak{P}$ is ramified.

Key Definitions

- **Dedekind’s discriminant theorem.** A rational prime p ramifies in K if and only if $p \mid d_K$. (It is d_K , not $\text{disc}(f)$.)
- **Order.** An order in K is a subring $\mathcal{O} \subseteq \mathcal{O}_K$ that is a free $\mathbb{Z}$ -module of rank n . Equivalently, it has finite index in $\mathcal{O}_K$. The maximal order is $\mathcal{O}_K$.
- **Conductor of an order.** $\mathfrak{f} = \{x \in \mathcal{O}_K : x\mathcal{O}_K \subseteq \mathcal{O}\}$. It is the largest ideal of $\mathcal{O}_K$ contained in $\mathcal{O}$. For example, $\mathbb{Z}[\sqrt{-3}] = \mathbb{Z} + 2\mathbb{Z}[\omega]$ has conductor $2\mathbb{Z}[\omega]$, where $\omega = \frac{1+\sqrt{-3}}{2}$.
- **Index prime.** A prime p dividing $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$. At such primes Dedekind–Kummer

applied to α may give the wrong answer.

Learning objectives

Students will be able to:

- **determine** the ramified primes of a field from d_K ;
- **diagnose** when Dedekind–Kummer with a given generator is unreliable, and **correct** it;
- **evaluate** whether an ideal in a non-maximal order is invertible;
- **explain** the chain “ramified $\iff$ nilpotents $\iff$ degenerate trace form $\iff p \mid d_K$ ”.

Prerequisites

- Discriminants and the index formula (earlier in this module: T3).
- Dedekind–Kummer (earlier in this module: T5).
- Invertibility of ideals (earlier in this module: T2).
- Nondegenerate bilinear forms over a field (assumed background).

Example questions with solutions

Q6.1 List the ramified primes of $\mathbb{Q}(\sqrt{-5})$, $\mathbb{Q}(\sqrt{13})$, $\mathbb{Q}(\sqrt[3]{2})$ (given $d_K = -108$) and $\mathbb{Q}(\zeta_5)$ (given $d_K = 125$).

Solution.

- $\mathbb{Q}(\sqrt{-5})$: $d_K = -20 = -2^2 \cdot 5$, so $\{2, 5\}$.
- $\mathbb{Q}(\sqrt{13})$: $d_K = 13$, so $\{13\}$.
- $\mathbb{Q}(\sqrt[3]{2})$: $-108 = -2^2 \cdot 3^3$, so $\{2, 3\}$. This matches Q5.2.
- $\mathbb{Q}(\zeta_5)$: $125 = 5^3$, so $\{5\}$. This matches Q5.3.

Q6.2 A student applies Dedekind–Kummer to $x^2 - 13$ at $p = 2$. They get $x^2 - 13 \equiv (x+1)^2 \pmod{2}$ and conclude that 2 ramifies in $\mathbb{Q}(\sqrt{13})$. Find the error and give the correct factorization of (2).

Solution.

- The error: $[\mathcal{O}_K : \mathbb{Z}[\sqrt{13}]] = 2$ (Q3.2), so 2 is an index prime and the theorem does not apply with $\alpha = \sqrt{13}$. Consistently, $2 \nmid d_K = 13$, so 2 cannot ramify.
- The correction: use $\theta = \frac{1+\sqrt{13}}{2}$, whose minimal polynomial is $x^2 - x - 3$. Mod 2 this is $x^2 + x + 1$, which has no roots in $\mathbb{F}_2$ and is therefore irreducible.
- So 2 is inert: (2) is prime with residue field $\mathbb{F}_4$. This matches the quadratic law, since $13 \equiv 5 \pmod{8}$.

Q6.3 In $R = \mathbb{Z}[\sqrt{-3}]$:

- (a) Show that $P = (2, 1 + \sqrt{-3})$ is the only prime of R containing 2.
- (b) Show that (2) is not a power of P .

(c) Compare with $\mathbb{Z}[\omega]$.

Solution.

- (a) $R/(2) \cong \mathbb{F}_2[x]/(x^2+3) = \mathbb{F}_2[x]/((x+1)^2)$. This ring has a unique prime ideal, generated by $x+1$. Pulled back to R , it is P , with $R/P \cong \mathbb{F}_2$.
- (b) $P \neq (2)$, since $1 + \sqrt{-3} \notin 2R$. For $k \geq 2$, $P^k \subseteq P^2 = 2P$ by Q2.3. Now $[R : 2P] = [R : 2R][2R : 2P] = 4 \cdot 2 = 8$, while $[R : 2R] = 4$. So $P^k \neq (2)$ for every k .
- (c) In $\mathbb{Z}[\omega]$ the minimal polynomial is $x^2 - x + 1 \equiv x^2 + x + 1 \pmod{2}$, which is irreducible. So 2 is inert with norm 4, and unique factorization is restored. The failure in R happens exactly at the conductor prime 2.

Instructor note — visual aid (sketch on the board)

The order $\mathbb{Z}[\sqrt{-3}]$ inside $\mathbb{Z}[\omega]$ (hexagonal lattice).

- Draw the complex plane and plot the points $a + b\omega$ for a, b from -2 to 2 , where ω sits at $(\frac{1}{2}, \frac{\sqrt{3}}{2}) \approx (0.5, 0.87)$. This gives a triangular (hexagonal) lattice; make these points hollow dots.
- Fill in the dots in rows with b even, that is, at heights 0 and $\pm\sqrt{3}$. These filled dots are $\mathbb{Z}[\sqrt{-3}]$.
- Label 1 , ω and $\sqrt{-3} = 2\omega - 1$.
- Circle one hollow dot, ω , and write: “integral over R , in $\text{Frac}(R)$, not in R ”.
- Beside the picture, write “index $2 \Rightarrow \text{disc}(R) = 2^2 \cdot (-3) = -12$ ”.
- The takeaway: the missing half-rows are exactly what integral closure adds back, and 2 is exactly where things break.

2.7 T7. Galois extensions: decomposition, inertia, Frobenius (Session 7)

Concepts

- For $K/\mathbb{Q}$ Galois with group G , G acts transitively on the primes above p . Hence all e_i are equal, all f_i are equal, and $efg = n$.
- $|D(\mathfrak{P} | p)| = ef$ (orbit–stabilizer: $g = [G : D]$) and $|I(\mathfrak{P} | p)| = e$.
- The exact sequence $1 \rightarrow I \rightarrow D \rightarrow \text{Gal}(k(\mathfrak{P})/\mathbb{F}_p) \rightarrow 1$, whose surjectivity is the key lemma. Here $\text{Gal}(k(\mathfrak{P})/\mathbb{F}_p)$ is cyclic, generated by $x \mapsto x^p$.
- The Frobenius element at an unramified $\mathfrak{P}$.
- Dependence on the choice of $\mathfrak{P}$: $D(\sigma\mathfrak{P} | p) = \sigma D(\mathfrak{P} | p) \sigma^{-1}$ and $\text{Frob}_{\sigma\mathfrak{P}} = \sigma \text{Frob}_{\mathfrak{P}} \sigma^{-1}$. So p determines a conjugacy class.
- The tower $\mathbb{Q} \subseteq K^D \subseteq K^I \subseteq K$. Going up, the steps carry g (splitting, with $e = f = 1$), then f (residue growth), then e (total ramification).
- Criteria: p is unramified $\iff I = 1$; p splits completely $\iff D = 1$.
- Cycle type: for $p \nmid \text{disc}(f)$, the cycle type of $\text{Frob}_{\mathfrak{P}}$ on the roots of f equals the list of degrees of the irreducible factors of $f \pmod{p}$.

- In the abelian case, D , I and Frob depend only on p .

Key Definitions

- **Decomposition group.** $D(\mathfrak{P} \mid p) = \{\sigma \in G : \sigma(\mathfrak{P}) = \mathfrak{P}\}$, the stabilizer of $\mathfrak{P}$. It depends on $\mathfrak{P}$, not just on p .
- **Inertia group.** $I(\mathfrak{P} \mid p) = \{\sigma \in D(\mathfrak{P} \mid p) : \sigma(x) \equiv x \pmod{\mathfrak{P}} \text{ for all } x \in \mathcal{O}_K\}$, the kernel of $D \rightarrow \text{Gal}(k(\mathfrak{P})/\mathbb{F}_p)$.
- **Frobenius element.** If p is unramified in K , then $\text{Frob}_{\mathfrak{P}}$ is the *unique* $\sigma \in D(\mathfrak{P} \mid p)$ such that $\sigma(x) \equiv x^p \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_K$. If p ramifies, this condition determines only a coset of I .
- **Decomposition and inertia fields.** K^D and K^I , the fixed fields of $D(\mathfrak{P} \mid p)$ and $I(\mathfrak{P} \mid p)$.

Learning objectives

Students will be able to:

- **derive** $efg = n$, $|D| = ef$ and $|I| = e$ from transitivity;
- **compute** D , I and Frob in quadratic, cyclotomic and S_3 examples;
- **explain** why these objects depend on $\mathfrak{P}$ only up to conjugacy;
- **identify** decomposition and inertia fields and verify the splitting behavior in each layer of the tower;
- **determine** the cycle type of Frob from a factorization mod p .

Prerequisites

- The fundamental theorem of Galois theory; Galois groups of finite fields (assumed background).
- e , f , g and the fundamental identity (earlier in this module: T4).
- Dedekind–Kummer and cyclotomic splitting (earlier in this module: T5).
- Ramified primes via d_K (earlier in this module: T6).

Example questions with solutions

Q7.1 $K = \mathbb{Q}(i)$, $G = \{1, c\}$ with c complex conjugation. Compute D , I and (where defined) Frob at $p = 2, 3, 5$.

Solution.

- $p = 5$ splits, so $g = 2$, $ef = 1$, and $D = I = 1$. Hence $\text{Frob} = 1$.
- $p = 3$ is inert, so $D = G$ ($ef = 2$) and $I = 1$ ($e = 1$). Hence $\text{Frob} = c$.
Direct check: $(a + bi)^3 \equiv a^3 + b^3 i^3 \equiv a^3 - b^3 i \equiv a - bi \pmod{3}$, by the binomial theorem mod 3 and Fermat.
- $p = 2$: $(2) = (1 + i)^2$, so $D = I = G$. The Frobenius is only defined as the coset $I = G$.

Q7.2 $K = \mathbb{Q}(\zeta_8)$, $G = \{\sigma_1, \sigma_3, \sigma_5, \sigma_7\}$ with $\sigma_a(\zeta_8) = \zeta_8^a$. For $p = 3$, find e, f, g, D, I , Frob and the decomposition field. Check the splitting in that field.

Solution.

- 3 is unramified, so $e = 1$. Also $\text{ord}_8(3) = 2$, so $f = 2$ and $g = 2$.
- Hence $D = \langle \text{Frob} \rangle$ has order 2 and $I = 1$. The Frobenius is σ_3 : in general $\text{Frob}_p = \sigma_p$, since $\sigma_p(\zeta) = \zeta^p$ and $x \mapsto x^p$ is additive mod p .
- Decomposition field: $\sqrt{-2} = \zeta + \zeta^3$ (check: $(\zeta + \zeta^3)^2 = \zeta^2 + 2\zeta^4 + \zeta^6 = i - 2 - i = -2$). Also $\sigma_3(\zeta + \zeta^3) = \zeta^3 + \zeta^9 = \zeta^3 + \zeta$. So $K^D = \mathbb{Q}(\sqrt{-2})$.
- Check: $-2 \equiv 1 \pmod{3}$ is a square, and indeed $3 = (1 + \sqrt{-2})(1 - \sqrt{-2})$ splits in $\mathbb{Q}(\sqrt{-2})$. Each of these two primes is then inert in K .

Q7.3 Let $L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, the splitting field of $x^3 - 2$, with $G \cong S_3$. For $p = 5$, find e, f and g , and describe the decomposition groups of the primes above 5.

Solution.

- The ramified primes of L lie among those of $\mathbb{Q}(\sqrt[3]{2})$ (namely 2 and 3) and $\mathbb{Q}(\zeta_3)$ (namely 3). So 5 is unramified: $e = 1$ and $I = 1$.
- By Q5.2, $x^3 - 2 \equiv (x - 3)(x^2 + 3x + 4) \pmod{5}$ with the quadratic irreducible. So the Frobenius fixes one root and swaps the other two: it is a transposition.
- Hence $f = |D| = 2$ and $g = 6/2 = 3$.
- The three primes above 5 have as decomposition groups the three distinct subgroups of order 2 (conjugate transpositions). No single subgroup is “the” decomposition group of 5.
- Consistency check: $5 \equiv 2 \pmod{3}$ is inert in $\mathbb{Q}(\zeta_3)$, which forces $2 \mid f$.

Instructor note — visual aid (sketch on the board)

(1) The D/I tower.

- Draw a vertical tower of four fields, bottom to top: $\mathbb{Q}, K^D, K^I, K$. Join consecutive fields by vertical line segments.
- Label the segment degrees on the left: g from $\mathbb{Q}$ to K^D , f from K^D to K^I , e from K^I to K .
- On the right, write the corresponding subgroups: G beside $\mathbb{Q}$, D beside K^D , I beside K^I , 1 beside K .
- Beside each segment, write what happens to the prime below $\mathfrak{P}$ there:
 - bottom: “splits off, $e = f = 1$ ”;
 - middle: “inert, residue field grows”;
 - top: “totally ramified”.
- Work the example $\mathbb{Q}(\zeta_{12})$ at $p = 3$ in a second column: $K^D = \mathbb{Q}$, $K^I = \mathbb{Q}(i)$, $K = \mathbb{Q}(\zeta_{12})$, with degrees 1, 2, 2.

(2) Three primes, three transpositions.

- Draw three dots in a row labeled $\mathfrak{P}_1, \mathfrak{P}_2, \mathfrak{P}_3$, all above one dot labeled 5.

- Under each $\mathfrak{P}_i$, write its decomposition group as a transposition acting on the roots $\{\beta_1, \beta_2, \beta_3\}$ of $x^3 - 2$: $(\beta_2 \beta_3)$, $(\beta_1 \beta_3)$, $(\beta_1 \beta_2)$.
- Draw curved arrows labeled by a 3-cycle τ moving $\mathfrak{P}_1 \rightarrow \mathfrak{P}_2 \rightarrow \mathfrak{P}_3$. Next to them, write $D(\tau\mathfrak{P}) = \tau D(\mathfrak{P})\tau^{-1}$.

2.8 T8 (optional). Frobenius in cyclotomic fields; quadratic reciprocity (Session 8)

Concepts

- In $\mathbb{Q}(\zeta_m)$, for $p \nmid m$: $\text{Frob}_p = \sigma_p$, where $\sigma_p(\zeta_m) = \zeta_m^p$. This recovers the cyclotomic splitting law, since $f = \text{ord}(\sigma_p) = \text{ord}_m(p)$.
- Splitting in subfields (abelian case). Let $F \subseteq L$ with $L/\mathbb{Q}$ abelian and p unramified in L . Then p splits completely in $F \iff \text{Frob}_p \in \text{Gal}(L/F)$.
- $\mathbb{Q}(\zeta_p)$ contains a unique quadratic subfield, $\mathbb{Q}(\sqrt{p^*})$, proved via a Gauss sum. Its Galois group corresponds to the squares in $(\mathbb{Z}/p)^\times$.
- Quadratic reciprocity: for odd primes $q \neq p$, $\left(\frac{p^*}{q}\right) = \left(\frac{q}{p}\right)$.
- The supplementary law for 2 via $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\zeta_8)$.

Key Definitions

- **Artin symbol (abelian, unramified case).** If $L/\mathbb{Q}$ is abelian and p is unramified, all $\text{Frob}_{\mathfrak{P}}$ with $\mathfrak{P} \mid p$ coincide. The common element is written Frob_p or $\left(\frac{L/\mathbb{Q}}{p}\right)$.
- **Signed prime.** For an odd prime p , $p^* = (-1)^{(p-1)/2}p$. It is the unique sign making $p^* \equiv 1 \pmod{4}$.
- **Quadratic Gauss sum.** $g_p = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^a$. It satisfies $g_p^2 = p^*$.

Learning objectives

Students will be able to:

- **compute** Frobenius elements in cyclotomic fields;
- **derive** splitting laws for subfields from the position of Frob_p ;
- **prove** quadratic reciprocity from the embedding $\mathbb{Q}(\sqrt{p^*}) \subseteq \mathbb{Q}(\zeta_p)$;
- **verify** the resulting laws numerically for small primes.

Prerequisites

- Frobenius elements and decomposition groups (earlier in this module: T7).
- Cyclotomic splitting (earlier in this module: T5).
- Quadratic splitting law (earlier in this module: T5).
- Legendre symbol and Euler's criterion (assumed background).

Example questions with solutions

Q8.1 Show that $\mathbb{Q}(\sqrt{5}) \subseteq \mathbb{Q}(\zeta_5)$. Deduce that a prime $q \neq 5$ splits in $\mathbb{Q}(\sqrt{5})$ if and only if $q \equiv \pm 1 \pmod{5}$. Check the cases $q = 11$ and $q = 7$.

Solution.

- $g_5 = \zeta - \zeta^2 - \zeta^3 + \zeta^4$ satisfies $g_5^2 = 5^* = 5$. So $\sqrt{5} \in \mathbb{Q}(\zeta_5)$.
- $\text{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}(\sqrt{5}))$ is the unique subgroup of order 2 in $(\mathbb{Z}/5)^\times$, namely $\{1, 4\} = \{\pm 1\}$.
- q splits in $\mathbb{Q}(\sqrt{5}) \iff \text{Frob}_q = \sigma_q \in \{\sigma_1, \sigma_4\} \iff q \equiv \pm 1 \pmod{5}$.
- $q = 11 \equiv 1$: splits. Direct check: $x^2 - x - 1$ has discriminant $5 \equiv 16 = 4^2 \pmod{11}$, so it has roots mod 11.
- $q = 7 \equiv 2$: inert. Direct check: the squares mod 7 are 1, 2, 4, so 5 is not a square.

Q8.2 Use $\mathbb{Q}(\sqrt{-7}) \subseteq \mathbb{Q}(\zeta_7)$ to show that, for $q \neq 7$, q splits in $\mathbb{Q}(\sqrt{-7})$ if and only if q is a square mod 7. Check $q = 2$, $q = 3$ and $q = 11$ against the quadratic splitting law.

Solution.

- $7^* = -7$, and the unique quadratic subfield of $\mathbb{Q}(\zeta_7)$ is $\mathbb{Q}(\sqrt{-7})$. Its Galois group is the index-2 subgroup $\{\sigma_1, \sigma_2, \sigma_4\}$, the squares mod 7.
- So q splits $\iff \sigma_q$ lies in that subgroup $\iff q \pmod{7} \in \{1, 2, 4\}$.
- $q = 2$: 2 is a square mod 7, so 2 splits. This agrees with the law at 2, since $-7 \equiv 1 \pmod{8}$.
- $q = 3$: 3 is not a square mod 7, so 3 is inert. This agrees with $-7 \equiv 2 \pmod{3}$ being a non-square.
- $q = 11$: $11 \equiv 4$ is a square mod 7, so 11 splits. This agrees with $\left(\frac{-7}{11}\right) = \left(\frac{4}{11}\right) = 1$.

In general, the two characterizations together give $\left(\frac{-7}{q}\right) = \left(\frac{q}{7}\right)$, which is reciprocity for $p = 7$.

Instructor note — visual aid (sketch on the board)

Subfield lattice of $\mathbb{Q}(\zeta_8)$ with splitting annotations.

- Put $\mathbb{Q}(\zeta_8)$ at the top and $\mathbb{Q}$ at the bottom.
- In the middle row, place $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. Join each to both the top and the bottom.
- Beside each middle field, write the subgroup of $(\mathbb{Z}/8)^\times$ fixing it: $\{1, 5\}$ for $\mathbb{Q}(i)$, $\{1, 7\}$ for $\mathbb{Q}(\sqrt{2})$, $\{1, 3\}$ for $\mathbb{Q}(\sqrt{-2})$.
- Under the diagram, write a small key relating $p \pmod{8}$ to the unique middle field in which p splits:
 - $3 \mapsto \mathbb{Q}(\sqrt{-2})$;
 - $5 \mapsto \mathbb{Q}(i)$;
 - $7 \mapsto \mathbb{Q}(\sqrt{2})$;
 - $1 \mapsto$ all three (split completely).
- Circle the $\mathbb{Q}(\sqrt{2})$ entry and note: “2 is a square mod $p \iff p \equiv \pm 1 \pmod{8}$ ” (the

supplementary law).

3 Difficult concepts

D1. “ $\mathcal{O}_K = \mathbb{Z}[\alpha]$ ” by default (T1, T5).

- *Misconception:* students assume $\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \mathbb{Z}[\sqrt{d}]$ always. More generally, they assume that any integral generator α of K gives $\mathcal{O}_K = \mathbb{Z}[\alpha]$.
- *Mitigation:* open Session 1 with $\varphi = \frac{1+\sqrt{5}}{2}$ and do the trace/norm derivation (Q1.3) live. Revisit the point in Session 6 with Q6.2, so that the mistake produces a visibly false conclusion (“2 ramifies in $\mathbb{Q}(\sqrt{13})$ ”).

D2. Direction of divisibility (T2).

- *Misconception:* writing $\mathfrak{p} \mid \mathfrak{a} \iff \mathfrak{p} \subseteq \mathfrak{a}$, the reverse of the truth, because “divides” feels like “is smaller”.
- *Mitigation:* anchor it in $\mathbb{Z}$: $2 \mid 6$ and $(2) \supseteq (6)$. Use the Hasse diagram of (6) in $\mathbb{Z}[\sqrt{-5}]$, with the rule “higher = bigger = divisor”. Have students compute $\gcd = \mathfrak{a} + \mathfrak{b}$ in one quick example before any proof.

D3. Inverses of fractional ideals (T2).

- *Misconception:* expecting $\mathfrak{p}^{-1} \subseteq \mathcal{O}_K$; conflating “invertible” with “principal”; trying to invert an ideal by inverting its generators one at a time.
- *Mitigation:* compute $\mathfrak{p}_2^{-1} = \frac{1}{2}\mathfrak{p}_2$ explicitly and exhibit $\frac{1+\sqrt{-5}}{2} \in \mathfrak{p}_2^{-1} \setminus \mathcal{O}$ (Q2.2). Then set side by side an ideal that is invertible but not principal ($\mathfrak{p}_2$ in $\mathbb{Z}[\sqrt{-5}]$) and one that is not invertible (P in $\mathbb{Z}[\sqrt{-3}]$, Q2.3).

D4. Irreducible elements versus prime ideals (T2).

- *Misconception:* believing that unique ideal factorization means an irreducible element generates a prime ideal.
- *Mitigation:* show that 2 is irreducible in $\mathbb{Z}[\sqrt{-5}]$ (no element has norm 2), yet $(2) = \mathfrak{p}_2^2$ is not prime. Say explicitly: “irreducible elements can generate non-prime ideals; the primes that divide (2) need not be principal.”

D5. Keeping e , f and g apart (T4).

- *Misconceptions:*
 - confusing the multiplicity e with the count g ;
 - reading “inert” as “nothing happens” instead of “the residue field grows to $\mathbb{F}_{p^n}$ ”;
 - assuming all e_i (or all f_i) are equal outside the Galois case.
- *Mitigation:* always write $\mathcal{O}_K/p\mathcal{O}_K$ as a product of local rings and read off e (nilpotency length) and f (residue field size) separately (Q4.2). Use the cubic splitting-type table (Q4.3), which includes the unequal type $\mathfrak{P}^2\mathfrak{q}$, and point to its concrete occurrence at 23 for $x^3 - x - 1$ in the question bank.

D6. Dedekind–Kummer at index primes (T5, T6).

- *Misconception:* applying the theorem to $x^2 - d$ at $p = 2$ when $d \equiv 1 \pmod{4}$, or reading ramification off $\text{disc}(f)$ rather than d_K .
- *Mitigation:* institute a checklist before any Dedekind–Kummer computation: “compute $\text{disc}(f)/d_K$; if p divides the index, change generator.” Run Q6.2 ($\mathbb{Q}(\sqrt{13})$ at 2) in two columns, wrong on the left and right on the right.

D7. Ramification $\iff p \mid d_K$: the abstraction jump (T6).

- *Misconception:* the discriminant is seen as “just a number”. The link between a determinant and nilpotents feels magical.
- *Mitigation:* work $\mathbb{Z}[i] \bmod 2$ explicitly.
 - The trace form in the basis $(1, i)$ has matrix $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$, which is the zero matrix mod 2, hence degenerate.
 - In $\mathbb{Z}[i]/(2)$, $(1 + i)^2 = 2i \equiv 0$, so $1 + i$ is nilpotent. Nilpotents always have trace 0, and here the nilpotent lies in the radical of the trace form.

D8. D , I and Frob depend on $\mathfrak{P}$ (T7).

- *Misconceptions:*
 - treating $D(\mathfrak{P} \mid p)$ as determined by p ;
 - treating “the Frobenius of p ” as a single element in a non-abelian extension;
 - using Frob at ramified primes.
- *Mitigation:* use Q7.3 ($x^3 - 2$ at $p = 5$): three primes, three different decomposition groups, and one conjugacy class of transpositions. At ramified primes, say every time: “only a coset of I ”.

D9. Mixing up the layers of the D/I tower (T7).

- *Misconception:* putting e at the bottom and g at the top, or mixing up the fact that the *smaller* group I corresponds to the *bigger* field K^I .
- *Mitigation:* use the tower diagram with the mnemonic “split, then grow, then ramify” reading bottom to top. Check it on $\mathbb{Q}(\zeta_{12})$ at 3, where each layer can be verified with a quadratic subfield.

4 Module question bank

Numerical questions, solvable without a calculator, spread evenly across topics. Each is tagged by difficulty and topic.

B1. [Easy — T1] Let $\theta = \frac{1+\sqrt{-7}}{2}$.

- Find the minimal polynomial of θ .
- Determine $\mathcal{O}_{\mathbb{Q}(\sqrt{-7})}$ and d_K .
- Is $\mathbb{Z}[\sqrt{-7}]$ a Dedekind domain?

Solution.

- (a) $\theta^2 = \frac{1+2\sqrt{-7}-7}{4} = \frac{-3+\sqrt{-7}}{2}$, so $\theta^2 - \theta = \frac{-3+\sqrt{-7}-1-\sqrt{-7}}{2} = -2$. The minimal polynomial is $x^2 - x + 2$.
- (b) $-7 \equiv 1 \pmod{4}$, so $\mathcal{O}_K = \mathbb{Z}[\theta]$ and $d_K = -7$.
- (c) No. θ lies in its fraction field, is integral, and is not in $\mathbb{Z}[\sqrt{-7}]$. So $\mathbb{Z}[\sqrt{-7}]$ is not integrally closed.

B2. [Easy — T2] In $\mathbb{Z}[\sqrt{-6}]$, let $\mathfrak{p}_2 = (2, \sqrt{-6})$ and $\mathfrak{p}_3 = (3, \sqrt{-6})$.

- (a) Show that $\mathfrak{p}_2^2 = (2)$, $\mathfrak{p}_3^2 = (3)$ and $\mathfrak{p}_2\mathfrak{p}_3 = (\sqrt{-6})$.
- (b) Use this to reconcile $6 = 2 \cdot 3 = -(\sqrt{-6})^2$.
- (c) Show that $\mathfrak{p}_2$ is not principal.

Solution.

- (a) $\mathfrak{p}_2^2 = (4, 2\sqrt{-6}, -6)$. It contains $2 = -6 + 2 \cdot 4$, and every generator is divisible by 2, so $\mathfrak{p}_2^2 = (2)$.
 $\mathfrak{p}_3^2 = (9, 3\sqrt{-6}, -6)$. It contains $3 = 9 - 6$, and every generator is divisible by 3, so $\mathfrak{p}_3^2 = (3)$.
 $\mathfrak{p}_2\mathfrak{p}_3 = (6, 2\sqrt{-6}, 3\sqrt{-6}, -6)$. It contains $\sqrt{-6} = 3\sqrt{-6} - 2\sqrt{-6}$. Conversely, every generator is a multiple of $\sqrt{-6}$, since $6 = -(\sqrt{-6})^2$. So $\mathfrak{p}_2\mathfrak{p}_3 = (\sqrt{-6})$.
- (b) Both factorizations give $(6) = \mathfrak{p}_2^2\mathfrak{p}_3^2$: the first as $(2)(3)$, the second as $(\sqrt{-6})^2$.
- (c) A generator would satisfy $a^2 + 6b^2 = 2$, which has no integer solutions.

B3. [Medium — T3] In $\mathbb{Z}[\sqrt{-5}]$, let $\mathfrak{q} = (7, 3 + \sqrt{-5})$ and $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$.

- (a) Compute $N(\mathfrak{q})$ and show that $\mathfrak{q}$ is prime.
- (b) Show that $\mathfrak{q}$ is not principal.
- (c) Find a generator of the principal ideal $\mathfrak{p}_2\mathfrak{q}$.

Solution.

- (a) $\mathbb{Z}[\sqrt{-5}]/\mathfrak{q} \cong \mathbb{Z}[x]/(7, x + 3, x^2 + 5)$. Setting $x = -3$ gives $9 + 5 = 14 \equiv 0 \pmod{7}$, so the quotient is $\mathbb{F}_7$. Hence $N(\mathfrak{q}) = 7$, and $\mathfrak{q}$ is prime (maximal) because the quotient is a field.
- (b) A generator would satisfy $a^2 + 5b^2 = 7$. For $b = 0$, 7 is not a square; for $b = \pm 1$, $a^2 = 2$ is impossible; $|b| \geq 2$ is too big. So $\mathfrak{q}$ is not principal.
- (c) $N(\mathfrak{p}_2\mathfrak{q}) = 14$, and $a^2 + 5b^2 = 14$ gives $3 \pm \sqrt{-5}$. Take $3 + \sqrt{-5}$:
- it lies in $\mathfrak{q}$ by definition;
 - it lies in $\mathfrak{p}_2 = \{a + b\sqrt{-5} : a \equiv b \pmod{2}\}$, since 3 and 1 are both odd.

So $(3 + \sqrt{-5}) \subseteq \mathfrak{p}_2 \cap \mathfrak{q} = \mathfrak{p}_2\mathfrak{q}$ (the two primes are coprime). Both sides have norm 14, so $\mathfrak{p}_2\mathfrak{q} = (3 + \sqrt{-5})$. This is also consistent with $h_K = 2$: the product of two nonprincipal ideals is principal.

B4. [Hard — T4/T6] Let α be a root of $f = x^3 - x - 1$. You may use that $\text{disc}(f) = -23$.

- (a) Explain why $\mathcal{O}_K = \mathbb{Z}[\alpha]$.
- (b) Factor 2, 5 and 23 in $\mathcal{O}_K$, giving e and f for each prime and checking $\sum e_i f_i = 3$.

Solution.

- (a) $-23 = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 d_K$ and 23 is squarefree, so the index is 1.
- (b) • $p = 2$: $\bar{f} = x^3 + x + 1$ has no root in $\mathbb{F}_2$ ($f(0) = f(1) = 1$), so it is irreducible. Thus 2 is inert, with $f = 3$.
- $p = 5$: $f(2) = 8 - 2 - 1 = 5 \equiv 0$. Dividing, $\bar{f} = (x - 2)(x^2 + 2x + 3)$. Check:

$$(x - 2)(x^2 + 2x + 3) = x^3 - x - 6 \equiv x^3 - x - 1 \pmod{5}.$$

The quadratic has discriminant $4 - 12 = -8 \equiv 2 \pmod{5}$, a non-square. So

$$(5) = (5, \alpha - 2)(5, \alpha^2 + 2\alpha + 3),$$

with residue degrees 1 and 2 and all $e = 1$. Check: $1 + 2 = 3$.

- $p = 23$: since $23 \mid d_K$, the prime 23 ramifies, so $\bar{f}$ has a repeated root r . From $f'(r) = 3r^2 - 1 = 0$ we get $r^2 \equiv 3^{-1} \equiv 8 \pmod{23}$, because $3 \cdot 8 = 24 \equiv 1$. Now $10^2 = 100 = 92 + 8$, so $r \equiv \pm 10$.
Test $r = 10$: $f(10) = 1000 - 10 - 1 = 989 = 23 \cdot 43 \equiv 0$. So 10 is the double root.
The roots sum to 0 (there is no x^2 term), so the third root is $s \equiv -20 \equiv 3$. Check: $f(3) = 27 - 3 - 1 = 23 \equiv 0$. So

$$(23) = (23, \alpha - 10)^2(23, \alpha - 3),$$

with $e = 2, f = 1$ and $e = 1, f = 1$. Check: $2 + 1 = 3$. The prime 23 ramifies but not totally.

B5. [Medium — T6] Let $K = \mathbb{Q}(\sqrt{17})$.

- (a) Find d_K and the ramified primes.
- (b) Factor (2) correctly, and explain why applying Dedekind–Kummer to $x^2 - 17$ gives the wrong answer.
- (c) Determine the splitting of 3 and 13.

Solution.

- (a) $17 \equiv 1 \pmod{4}$, so $\mathcal{O}_K = \mathbb{Z}[\theta]$ with $\theta = \frac{1+\sqrt{17}}{2}$, and $d_K = 17$. Only 17 ramifies.
- (b) θ has minimal polynomial $x^2 - x - 4 \equiv x(x+1) \pmod{2}$. So $(2) = (2, \theta)(2, \theta + 1)$ splits, consistent with $17 \equiv 1 \pmod{8}$.
By contrast, $x^2 - 17 \equiv (x+1)^2 \pmod{2}$ would suggest ramification. But $[\mathcal{O}_K : \mathbb{Z}[\sqrt{17}]] = 2$, so Dedekind–Kummer does not apply to $\sqrt{17}$ at $p = 2$.
- (c) • $p = 3$: $17 \equiv 2 \pmod{3}$ is not a square, so 3 is inert. Directly: $x^2 - x - 4 \equiv x^2 + 2x + 2 \pmod{3}$ takes the values 2, 2, 1 at 0, 1, 2, so it has no roots.
- $p = 13$: $17 \equiv 4 = 2^2 \pmod{13}$, so 13 splits.

B6. [Easy — T5] In $\mathbb{Q}(\zeta_7)$ (with $\mathcal{O}_K = \mathbb{Z}[\zeta_7]$), determine e , f and g for $p = 2, 13, 29, 7$. For $p = 2$, give the prime ideals explicitly.

Solution.

- $p = 2$: $2^3 = 8 \equiv 1 \pmod{7}$, so $f = 3$ and $g = 2$. Explicitly,

$$\Phi_7 \equiv (x^3 + x + 1)(x^3 + x^2 + 1) \pmod{2}.$$

Expanding the product gives $x^6 + x^5 + x^4 + 3x^3 + x^2 + x + 1 \equiv \Phi_7$. So $(2) = (2, \zeta^3 + \zeta + 1)(2, \zeta^3 + \zeta^2 + 1)$.

- $p = 13 \equiv 6 \pmod{7}$: the order is 2, so $f = 2$ and $g = 3$.
- $p = 29 \equiv 1 \pmod{7}$: $f = 1$ and $g = 6$, so 29 splits completely.
- $p = 7$: totally ramified, $(7) = (1 - \zeta_7)^6$, with $e = 6$.

B7. [Medium — T7] In $K = \mathbb{Q}(\zeta_8)$, for $p = 5$, $p = 7$ and $p = 2$, compute e , f , g , D , I , Frob (where defined) and the decomposition field. Check the splitting in the decomposition field.

Solution.

- $p = 5$: $\text{ord}_8(5) = 2$, so $e = 1$, $f = 2$, $g = 2$. Then $D = \{\sigma_1, \sigma_5\}$, $I = 1$ and $\text{Frob} = \sigma_5$. Since $\sigma_5(i) = \zeta^{10} = \zeta^2 = i$, we get $K^D = \mathbb{Q}(i)$. Check: $5 = (2 + i)(2 - i)$ splits in $\mathbb{Q}(i)$.
- $p = 7$: $\text{ord}_8(7) = 2$, so again $e = 1$, $f = 2$, $g = 2$. Then $D = \{\sigma_1, \sigma_7\}$ and $\text{Frob} = \sigma_7$, which is complex conjugation ($\zeta^7 = \bar{\zeta}$). So $K^D = \mathbb{Q}(\zeta + \bar{\zeta}) = \mathbb{Q}(\sqrt{2})$. Check: $7 = (3 + \sqrt{2})(3 - \sqrt{2})$ splits in $\mathbb{Q}(\sqrt{2})$.
- $p = 2$: $\Phi_8 = x^4 + 1 \equiv (x + 1)^4 \pmod{2}$, so 2 is totally ramified with $e = 4$. Then $D = I = G$, the decomposition field is $\mathbb{Q}$, and the Frobenius is only defined as the coset $I = G$.

B8. [Hard — T7/T8] Let $K = \mathbb{Q}(\zeta_{12})$ with $G = \{\sigma_1, \sigma_5, \sigma_7, \sigma_{11}\}$, and quadratic subfields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{3})$.

- Identify which σ_a fixes each quadratic subfield.
- For $p = 3$ and $p = 2$, find e , f , g , D , I , the inertia field, and the coset of Frobenius elements.

Solution.

- Use $i = \zeta_{12}^3$ and $\zeta_3 = \zeta_{12}^4$.
 - σ_5 : $\zeta^{15} = \zeta^3$, so σ_5 fixes i and hence $\mathbb{Q}(i)$.
 - σ_7 : $\zeta^{28} = \zeta^4$, so σ_7 fixes ζ_3 and hence $\mathbb{Q}(\sqrt{-3})$.
 - σ_{11} is complex conjugation, which fixes the real subfield $\mathbb{Q}(\sqrt{3})$.
- $p = 3$: 3 is totally ramified in $\mathbb{Q}(\zeta_3)$, so $e = 2$. The part prime to 3 is $\mathbb{Q}(\zeta_4)$, and $\text{ord}_4(3) = 2$, so $f = 2$. Then $g = 4/(2 \cdot 2) = 1$. $D = G$, and $I = \{\sigma_1, \sigma_5\}$ (the elements $\equiv 1 \pmod{4}$). The inertia field is $\mathbb{Q}(i)$: 3 is unramified there, and in fact inert, which accounts for $f = 2$. The Frobenius coset is $\{\sigma_7, \sigma_{11}\}$. Both act on $\mathbb{Q}(i)$ as conjugation, as they should since $3 \equiv 3 \pmod{4}$.
 - $p = 2$: 2 is ramified in $\mathbb{Q}(\zeta_4)$, so $e = \varphi(4) = 2$. Also $\text{ord}_3(2) = 2$, so $f = 2$, and $g = 1$. $I = \{\sigma_1, \sigma_7\}$ (the elements $\equiv 1 \pmod{3}$), and the inertia field is $\mathbb{Q}(\sqrt{-3})$. Check: 2 is inert in $\mathbb{Q}(\sqrt{-3})$ because $-3 \equiv 5 \pmod{8}$, and it ramifies in both $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{3})$. The Frobenius coset is $\{\sigma_5, \sigma_{11}\}$.